

Annotation Convention, Not Architecture, Limits Cross-Facility Detection of Electric Vehicle Battery Components

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Abstract—Automated disassembly of end-of-life electric vehicle (EV) battery packs depends on a detector that locates modules and busbars across the pack types a recycling line actually receives. Published detectors for this task are trained and validated at a single facility. This paper assembles a cross-facility corpus from 13 independent public sources spanning ten pack families, verified free of train-test leakage by perceptual hashing, and disaggregates accuracy by source. Error proves concentrated rather than spread: per-source module accuracy spans 0.097 to 1.000, and the collapse occurs where a source disagrees about what the target class denotes rather than where the imagery is difficult. This is developed into a screening procedure that identifies convention-divergent sources from a single training run using a robust outlier criterion, and into a training-free variant computed from annotation geometry alone over 4,760 label files. Both pass a positive control in which a consensus source relabelled to a sub-component convention is recovered while no other source changes status. Two further results concern method rather than data. Under a matched 60-epoch budget, self-supervised pretraining is marginally worse on conventionally annotated packs, at 0.955 against 0.979, and better by a factor of 3.5 under annotation shift, at 0.337 against 0.097. And validation data drawn from a single divergent source is shown to invert checkpoint selection rather than merely bias it: the selected checkpoint is worse than the fixed-budget final checkpoint by 51% relative, with non-overlapping bootstrap intervals, in both architectures tested. Five negative results are reported. The corpus definition and evaluation code are released.

Index Terms—object detection, dataset quality, annotation consistency, domain generalisation, cross-dataset evaluation, electric vehicle batteries, circular economy

I. Introduction

Electric vehicles are reaching end of life in large numbers, and the packs they carry are simultaneously a hazard and a source of critical raw materials. Lithium, cobalt and nickel are all classified by the European Commission as critical raw materials with elevated supply risk [1], [2], and European recycling capacity demand is projected to reach 170 GWh by 2035 [3]. Disassembly of these packs remains predominantly manual. The operation is labour intensive, slow, and exposes workers to high voltage and to the risk

of thermal runaway, which motivates a substantial and growing research effort on robotic disassembly [4], [6].

Every such system begins with perception. Before a manipulator can unfasten, grasp or lift a component, the battery modules and the busbars that connect them must be located in the image. The established approach is a supervised object detector trained on imagery collected at one facility or from one experimental rig. This practice conceals a problem specific to the application. A recycling line does not receive a single pack type; it receives whatever arrives on a given day, from different manufacturers, in different states of disassembly, under whatever illumination the building provides. Accuracy measured on the distribution a detector was trained on therefore carries limited information about the next pack through the door.

To quantify this, a cross-facility benchmark for EV battery component detection is required, and none exists. Every study surveyed in Section II collected and annotated its own imagery, and none released it. To address this gap, this paper assembles a corpus of 4,425 training images from 13 independent public sources spanning ten EV pack families, constructs a held-out cross-facility benchmark of 225 images and 780 instances, and evaluates detectors on it under a single harmonised protocol.

The principal finding is that the dominant failure mode is not image difficulty and not model capacity. It is disagreement between sources about what the target class denotes. Per-source module accuracy spans 1.000 to 0.097, and inspection confirms that the collapsing source annotates small internal sub-components as modules while every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently; the collapse is a disagreement about the target, not a failure of perception.

This paper makes the following contributions:

- 1) A cross-facility corpus and benchmark for EV battery component detection, assembled from 13 public sources over ten pack families, leakage-checked and released with per-source provenance and licences (Section III).
- 2) A convention-consistency screening procedure identifying annotation-divergent sources from a single

training run, with a positive control in which a relabelled consensus source is recovered and no other source changes status (Sections IV-C, V-B and V-D).

- 3) A training-free variant computed from annotation geometry alone, which identifies the same source with no training run (Sections IV-D and V-C).
- 4) A matched-budget architecture comparison showing self-supervised pretraining to be marginally worse on conventionally annotated packs and better by a factor of 3.5 under annotation shift (Section V-E).
- 5) The finding that single-source validation inverts checkpoint selection, costing 51% relative accuracy, in both architectures tested (Section V-F).
- 6) Five negative results constraining the design space (Section V-H).

Relation to prior work by the same authors. The single-source specialist evaluated here is the detector of [18], which addressed detection and condition assessment at one facility. That work could not test cross-facility transfer, which is the limitation this paper addresses. The corpus, the benchmark, both screening procedures and all evaluation reported here are new.

II. Related Work

A. Perception for battery disassembly

Zang et al. [4] surveyed robotic EV battery disassembly and concluded that the process remains predominantly manual, with perception and task uncertainty identified as the principal obstacles. Wegener et al. [5] demonstrated that pack designs vary sufficiently between manufacturers that fixed automation does not transfer, which is the practical reason perception must generalise. However, that study addressed mechanical disassembly sequencing and did not evaluate a perception component.

Most published perception work for this application targets fasteners rather than components. Screw detection has been addressed with Faster R-CNN combined with rotation edge similarity [7], with YOLOX-family detectors [8], and in a recent comparison of YOLOv11 against Faster R-CNN [9]. However, each of these studies reports accuracy on imagery from a single collection and therefore cannot establish transfer. Module-level and busbar-level perception is comparatively under-explored. Tan et al. [6] developed a collaborative robot that unfastens and collects screws by fusing vision with force and torque sensing, reporting above 90% unfastening efficiency; however, the vision component operates on a single fixed pack geometry. The closest prior work applies instance segmentation with point-cloud registration to grasp busbars [10], but is restricted to one module type.

A recurring theme is data scarcity. Every study named above collected and hand-annotated its own imagery, 1,719 images in the case of [8], and none released it. [9] resorts to laptop screws as a proxy domain and states that the field is hampered by a lack of public datasets. That absence motivates the benchmark presented here.

B. Cross-dataset evaluation and label quality

Domain shift is a well-documented failure mode in visual recognition [11], [12]: models that appear strong on their own test split degrade sharply on data collected under a different protocol. However, these benchmarks vary the imaging distribution while holding the annotation protocol fixed, whereas the corpus considered here varies both. Northcutt et al. [13] demonstrated that pervasive label errors in established test sets destabilise benchmark rankings, but treated label error as noise about an agreed target rather than as systematic disagreement about what the target is. The distinction matters for merged corpora, and it is the distinction the proposed screening procedure operationalises. Penquitt et al. [22] developed a modular framework combining automatic label-error detection with crowd-sourced verification, recovering 18% missing or inaccurate labels in the KITTI pedestrian class. However, that framework corrects errors relative to a standard agreed within a single dataset, whereas the sources considered here disagree about what the class denotes, which no within-convention correction can reconcile. Abou Akar et al. [23] studied multi-source training for industrial detection using real, rendered and generatively synthesised imagery. However, those sources share one annotation standard by construction, since synthetic data carries its ground truth.

C. Self-supervised representations for detection

Benchmark design for industrial vision has precedent: MVTec AD [19] established that a carefully constructed benchmark can redirect a sub-field, though it was collected by one group under controlled conditions and so exhibits no inter-source annotation disagreement. Zhou et al. [21] surveyed domain generalisation; however, that taxonomy presumes a fixed label space across domains, the assumption shown here to fail on assembled corpora. Self-supervised backbones such as DINOv2 [24] learn features that transfer without task-specific fine-tuning, and detection transformers [20] built on them [25] are reported to generalise better than detectors trained from scratch in low-data regimes. However, that claim has been established on natural imagery with consistent annotation. Whether it holds on the cluttered, specular imagery of battery internals, and under annotation shift, is evaluated in Section V-E.

To the best of our knowledge, no public cross-facility benchmark for EV battery component detection exists, and no prior study has quantified the contribution of annotation-convention divergence to cross-source detection accuracy. Therefore, this research presents both.

III. Corpus and Benchmark

A. Sources

A corpus of 4,425 training images was assembled from 13 independent public sources spanning ten EV pack families, annotated with two classes, module and busbar (Table I).

TABLE I
Corpus sources (13 total, ten pack families)

Source	Images	Pack family
ev-battery-comp-det.	1,045	multi-manufacturer
ev-battery-components	680	Audi / BMW / Chevrolet
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single laboratory rig
battery_comp, ue_d1	517	Nissan Leaf, close-ups
Others (eight sources)	1,265	mixed

The images and the polygon annotations originate with those sources; the contribution made here is the assembly, the deduplication, the audit and the benchmark construction, not the annotation. Licences differ across sources and one is non-commercial, so per-source provenance is recorded and released so that any single source can be excluded by a downstream user.

B. Deduplication and leakage control

Reported accuracy is informative only against a benchmark that does not overlap the training data. All 4,440 candidate images were therefore compared by difference hashing (dHash) at a Hamming distance threshold of 6, a value selected because it separates augmentation-derived near-duplicates from genuinely distinct frames on this corpus. Fifteen same-source near-duplicates were removed, and a cross-source pass at a looser threshold returned zero duplicates, which confirms that the sources are genuinely independent rather than republications of one another.

This control proved necessary rather than precautionary. One candidate source, a 712-image public collection covering 19 vehicle types, republishes imagery from another source already present in the corpus: 78 images were byte-identical duplicates of training images. Merging without verification would have placed training images directly into the benchmark.

C. Annotation audit

All 4,425 training images were audited for label and image quality. Of these, 23.5% carry no labels, 12.3% score low on a Laplacian-variance blur metric, 20.4% are greyscale, and none contain degenerate boxes.

Two findings changed how the corpus was treated. Firstly, the blur and greyscale flags are largely false alarms. The lowest-scoring images are smooth, closed pack lids, for which a Laplacian-variance metric reads flat metal as blurred, and the greyscale images originate from two monochrome sources. Both add useful variation, so deletion on the metric alone would degrade the corpus. Secondly, the harmful category is unlabelled images that nonetheless contain visible components. A proportion of the 23.5% is legitimate background, which suppresses false positives, but inspection confirmed open packs with clearly visible busbar assemblies carrying no annotations. Such frames teach the detector that busbars are background and

are a plausible contributor to the weak busbar accuracy reported in Section V-E.

The audit also established that 16,945 of the 19,522 annotations are polygon masks with between 5 and 151 vertices rather than boxes. These are converted to enclosing boxes for detection training and evaluation. It should be noted that an early version of the evaluation pipeline silently discarded every polygon-format label, trained on 20% of the annotations, and produced an inflated score that appeared plausible. The failure is easy to introduce and difficult to detect, and ground-truth instance counts should be reconciled across pipelines as a matter of routine.

D. Evaluation sets

Two evaluation sets are used and kept distinct. A cross-facility benchmark of 225 images containing 780 instances was drawn from all sources and held out entirely from training; it supports the single-source comparison of Section V-A and nothing else. The screening procedures and every result depending on them operate on a stratified sample of 1,628 images spanning 11 sources drawn from the corpus itself, for the reason given in Section IV-C.

IV. Method

A. Detectors and training configurations

Three configurations are compared.

Single-source specialist. YOLOv8n trained only on the single-laboratory source at 768 px, following the conventional approach in this literature.

Multi-source YOLO11n. YOLO11n [26], 2.6 M parameters, trained from COCO initialisation [14] on all 13 sources at 640 px with SGD.

Multi-source RF-DETR. RF-DETR-Nano [25], a real-time detection transformer [20] initialised from the published checkpoint, whose backbone derives from self-supervised DINOv2 pretraining [24], fine-tuned end to end at 672 px. All 30.6 M parameters are updated; the backbone is not frozen.

Matched budget. An architecture comparison is meaningful only under equivalent training. Both multi-source models are therefore trained for exactly 60 epochs at the closest resolutions the architectures accept, RF-DETR requiring a multiple of 56. Early stopping is disabled and the final checkpoint taken, for the reason established in Section V-F. YOLO11n was trained with three seeds; RF-DETR once at the full budget, a limitation stated in Section VI.

B. Harmonised evaluation protocol

Confidence scores are not comparable across architectures, and mixing metric implementations produced a spurious result in early experiments conducted for this study. Every model is therefore scored with a single metric implementation at a common confidence threshold of 0.05, on identical images, against identical ground truth

including polygon-derived boxes. The threshold of 0.05 was selected because it lies below the operating point of all three detectors and therefore does not truncate the precision-recall curve for any of them.

C. Convention-consistency screening

The proposed screening procedure identifies sources whose annotation convention diverges from the corpus consensus, using a single merged-corpus training run.

Let $\mathcal{S} = \{S_1, \dots, S_n\}$ denote the sources and let a_i denote the class-specific mAP@50 obtained by the merged-corpus detector on the held-out images of S_i . A source is flagged as convention-divergent when its accuracy falls below a robust lower fence,

$$a_i < \text{med}(\mathbf{a}) - k \cdot 1.4826 \cdot \text{mad}(\mathbf{a}), \quad (1)$$

where $\mathbf{a} = (a_1, \dots, a_n)$, med denotes the median, mad denotes the median absolute deviation, the constant 1.4826 rescales the median absolute deviation to a consistent estimator of the standard deviation under normality, and k is a sensitivity parameter. A value of $k = 3$ is used throughout, which is the conventional choice for outlier rejection and which is shown in Section V-B to separate the divergent source from the consensus by a wide margin on this corpus.

The median and the median absolute deviation are used in place of the mean and the standard deviation because a divergent source inflates both of the latter and thereby masks itself. The procedure requires no additional annotation, no additional training run beyond the merged-corpus model that is trained in any case, and no assumption about which source is divergent.

Two conditions apply. A source must contribute at least 100 module instances to be screened, below which it is excluded rather than reported; at 40 instances an earlier evaluation produced a 0.36 mAP@50 swing on the smallest source between models differing only in an unrelated source’s labels. And the screen is computed on a stratified sample of the corpus rather than held-out data, because it is a curation diagnostic: a source contradicting the majority convention is not fitted well even in training. Absolute values are consequently optimistic and only orderings are used.

A flagged source is not discarded from the corpus. It is reported separately as an out-of-convention benchmark, on the grounds established in Section V-B that its labels encode a different task rather than a degraded version of the same one.

D. Training-free screening from annotation geometry

The procedure above requires a merged-corpus training run. A cheaper variant operates on the annotation files alone. A source annotating sub-components rather than pack-level modules produces many small boxes per image, whereas a source following the consensus convention

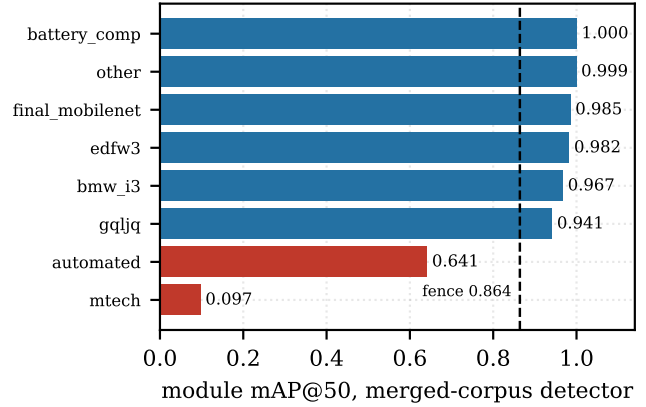


Fig. 1. Per-source module accuracy with the robust fence of Eq. 1 at $k = 3$. Two sources fall below it, one by an order of magnitude.

produces few large ones. That signature is captured by a granularity index, defined as follows:

$$g_i = n_i / s_i, \quad (2)$$

where n_i denotes the mean number of module instances per image in source S_i and s_i denotes the median normalised module scale, taken as the square root of the enclosing-box area in normalised image coordinates. Scale enters as a square root so that the index is a count per unit length and is invariant to image resolution. Eq. 1 is applied to $\mathbf{g}$ with the inequality reversed, since divergence raises the index.

V. Results

A. The single-source evaluation gap

The specialist attains 0.818 mAP@50 on the single-source benchmark it was trained and validated on, and 0.277 on a cross-facility benchmark drawn from all sources, a relative loss of 66%. It should be noted that the in-domain benchmark is itself the source shown below to diverge from the corpus consensus, so this figure conflates generalisation loss with training on an idiosyncratic convention and is read as an upper bound. No other result depends on it.

B. Per-source screening

The merged-corpus detector was evaluated per source on a stratified sample of 1,628 images spanning 11 sources. Fig. 1 presents the outcome: six sources lie between 0.941 and 1.000, one falls to 0.641 and one collapses to 0.097. Applying Eq. 1 at $k = 3$ gives a fence of 0.864, and two sources fall below it.

The two differ in character. The source at 0.097 lies 0.767 below the fence. Inspection confirms the diagnosis: it annotates small internal sub-components in extreme macro views as modules, whereas every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the

collapse is a disagreement about the target, not a failure of perception. The source at 0.641 is a milder case whose granularity index is the second highest in the corpus but does not exceed the training-free fence, so the two screens disagree about it and the model-based screen is the more sensitive. Three sources were excluded for contributing fewer than 100 module instances; one of them annotates busbars exclusively, which the screen surfaces without manual inspection.

C. Agreement of the training-free screen

Eq. 2 was evaluated over all 4,760 label files, using no images and no trained model. The divergent source attains 40.7, against 15.0 for the next highest and a corpus median of 9.00; the fence at $k = 3$ is 29.02 and exactly one source exceeds it. It is the same source the model-based procedure flags most strongly. The two screens are methodologically independent, one measuring detector behaviour and the other only annotation geometry, so their agreement is a second line of evidence that the source encodes a different convention rather than harder imagery. A corpus may therefore be screened before a detector is trained. The index detects divergence in annotation granularity specifically; conventions disagreeing about class boundaries while preserving object scale would still require the model-based screen.

D. Positive control

Neither screen had been tested against a case whose answer was known in advance. A consensus source was therefore relabelled to a deliberately different convention, every pack-level module box being replaced by a 3×2 grid of inset sub-component boxes covering the same region. No image is altered and no busbar annotation touched, so a screen responding to the manipulation responds to annotation convention alone. A detector was retrained on the manipulated corpus under the identical budget.

Table II presents the outcome. The manipulated source falls from 0.941 to 0.667 and crosses the fence, while no unmanipulated source moves by more than 0.050 and all but one move by 0.017 or less. The flagged set gains exactly one member and loses none. On the training-free index the same source rises from 11.1 to 185.8. Both screens therefore recover a divergence whose ground truth is known by construction, without false positives elsewhere.

E. Architecture comparison under a matched budget

Both multi-source models were trained for 60 epochs and scored per source under the identical protocol (Table III). The result is narrow and specific. On conventionally annotated packs the self-supervised pretrained model is marginally worse, at 0.955 against 0.979. On the mild outlier it scores 0.779 against 0.641, and on the divergent source 0.337 against 0.097, a factor of 3.5. Self-supervised pretraining therefore does not detect ordinary modules more accurately here; it degrades considerably

TABLE II
Positive control. A consensus source is relabelled to a sub-component convention and the detector retrained. Module mAP@50 above, granularity index below

Source	Original	Manipulated	Change
model-based screen, fence 0.864 / 0.815			
battery_comp	1.000	1.000	0.000
edfw3	0.982	0.973	-0.009
bmw_i3	0.967	0.950	-0.017
manipulated	0.941	0.667	-0.274
automated	0.641	0.591	-0.050
divergent	0.097	0.097	0.000
training-free screen, fence 29.02 / 35.24			
manipulated	11.1	185.8	+174.7

TABLE III
Architecture comparison under a matched 60-epoch budget, module mAP@50. Consensus is the mean over the six sources above the fence

Model	Consensus	Mild outlier	Divergent
YOLO11n	0.979	0.641	0.097
RF-DETR	0.955	0.779	0.337

more gracefully when the definition of the target moves. Both architectures produce the same partition under the screen, which indicates the procedure to be independent of the detector it is applied to.

F. Single-source validation inverts checkpoint selection

A result emerged from the matched-budget runs that was not sought and bears directly on the paper’s central claim. The validation data available for this corpus is drawn overwhelmingly from the divergent source. During training, accuracy on that split peaked early then fell by roughly a factor of five, while both training losses decreased monotonically. Selecting a checkpoint on that signal is therefore selecting for the divergent convention.

Evaluated across all sources, the selected checkpoint is worse than the final checkpoint by 0.235 mAP@50, a relative loss of 51%. Across three seeds the final checkpoint attains 0.696 ± 0.003 and the selected checkpoint 0.461 ± 0.018 , so the effect is roughly 80 times the seed spread. On the module class, 500 bootstrap resamples give non-overlapping 95% intervals of $[0.659, 0.808]$ and $[0.471, 0.626]$ (Fig. 2). The same inversion appears in the pretrained model, whose validation identified epoch 3 as best and epoch 60 as substantially worse, while across sources the epoch-60 checkpoint scores 0.955 against 0.719 on consensus sources and 0.337 against 0.079 on the divergent source. The effect is a property of single-source validation, not of one architecture.

Two caveats bound the claim. The evaluation sample is drawn from the corpus rather than held out, so later checkpoints are favoured by memorisation and the magnitude is optimistic; what memorisation does not explain

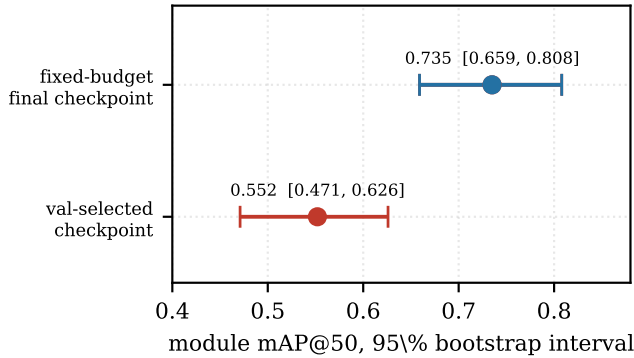


Fig. 2. Checkpoint selection against single-source validation data. Bars are 95% bootstrap intervals over 500 resamples; they do not overlap.

is the pattern, since memorising would raise the final checkpoint uniformly rather than gaining on consensus sources while losing on the divergent one. And the finding concerns validation drawn from a single divergent source, not well-constructed validation splits.

G. Deployment cost and zero-shot transfer

Evaluated on 17 pack families absent from the corpus, drawn from an unlabelled public collection, the merged-corpus detector attains a mean detection rate of 0.76 and finds busbars in 16 of 17, with 1.00 on BMW i4 and Hyundai Ioniq against 0.27 on a Volvo truck pack. Module geometry transfers to unseen passenger-vehicle packs while unusual form factors remain the weak axis. Dynamic INT8 quantisation to ONNX reduced median CPU latency from 56.3 ms to 44.8 ms and file size from 6.3 MB to 3.4 MB for a loss of 0.007 mAP@50, measured over 20 images on an Apple M1 processor, giving roughly 22 frames per second without a graphics processor.

H. Negative results

Five failures constrain the design space, each reported because it was expected to succeed. Open-vocabulary detection is unusable: YOLO-World [15], prompted with *ev battery module* and *busbar*, attained 0.004 mAP@50. Naive multi-source merging reduced module mAP@50 to 0.094 against a single-source baseline of 0.818, so more data made the model worse; this motivated the screening procedure. Automatic relabelling of the flagged source with Grounding DINO [16] recovered two of approximately five modules on clean frames and produced boxes covering half the frame on greyscale macro imagery, so convention conflicts are not cheaply repairable after collection. Copy-paste compositing [17] with synthetic glare degraded validation mAP@50 monotonically from 0.503 to 0.270 over six epochs. And prolonged fine-tuning degrades the pretrained model, which peaked at epoch 3 or 4 in both runs and ended 27% below its peak: a model arriving

already fitted gains nothing from a budget suited to one trained from scratch.

VI. Limitations

The 66% loss of Section V-A is measured against an in-domain benchmark that is itself the convention-divergent source, so it conflates generalisation loss with training on an idiosyncratic convention and is an upper bound. No other result depends on it.

Both screens are computed on a stratified sample of the corpus rather than held-out data, because the validation split available here spans only two sources and cannot support a per-source outlier test. Absolute values are optimistic and later checkpoints are additionally favoured by memorisation; only relative orderings are used. A held-out multi-source split would give cleaner magnitudes.

YOLO11n was trained with three seeds, giving a standard deviation of 0.003. The pretrained model was trained once at the full budget for reasons of compute; a second run reproduced its early-peak behaviour but was halted at epoch 9, so its numbers carry no variance estimate.

The screening procedure requires divergent sources to be a minority, since the median defines the consensus, and requires a minimum instance count per source; three sources here fall below that floor and are excluded rather than reported. It would need modification for a corpus in which two or more incompatible conventions are each well represented.

The benchmark inherits its sources' annotation conventions and no independently relabelled gold subset was constructed, so model error and label error cannot be separated on it. This is material for a study whose central claim concerns annotation quality. The corpus is assembled from public sources, so coverage is uneven and licences vary; one source at 24% of training images is CC BY-NC-SA 4.0 with a propagating ShareAlike condition, and one further source became unavailable during this study. Finally, busbar accuracy is not sufficient for autonomous operation, and these results characterise the failure modes of the task rather than deliver a deployable detector.

VII. Conclusion

This paper presented a cross-facility corpus and benchmark for electric vehicle battery component detection, and used it to establish that annotation convention, rather than architecture, is the dominant failure mode in this application. Per-source accuracy for one detector spans 0.097 to 1.000, and the collapse occurs where a source disagrees about what the target class denotes rather than where the imagery is hard. The proposed screening procedure identifies such sources from a single training run, and a training-free variant computed from annotation geometry identifies the same source with no training at all. Both pass a positive control in which a relabelled consensus source is recovered while no other source changes status.

Two results about method follow. Under a matched budget, self-supervised pretraining is marginally worse on conventionally annotated packs and better by a factor of 3.5 under annotation shift, which narrows the circumstances in which it should be preferred to graceful degradation rather than accuracy. And validation data drawn from a single divergent source inverts checkpoint selection, costing 51% relative accuracy, in both architectures tested.

Dataset design, and specifically agreement on annotation convention before collection, is therefore the first thing to get right in this application, and the same disagreement propagates silently into model selection if the validation split is not itself checked. Future work will construct an independently relabelled gold subset in order to separate model error from label error, and will repeat both screens on a held-out multi-source split.

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